Name of Role: Senior ML for Robotics Engineer – Edge AI Systems
Location: Abu Dhabi, UAE

Role Overview

Our client is seeking a **Senior ML for Robotics Engineer** to lead the architecture, deployment, and optimization of machine learning systems powering advanced autonomous robots, with a strong emphasis on humanoid platforms. This role owns the end-to-end ML stack, from model design and optimization pipelines to deployment and validation on edge hardware platforms, primarily NVIDIA Jetson devices.
The ideal candidate combines deep expertise in both traditional machine learning and modern AI techniques with strong systems-level thinking and embedded deployment experience. Working closely with robotics, infrastructure, and software teams, the Senior ML for Robotics Engineer will drive technical decisions that enable robust, real-time AI capabilities on physical robotic systems.

Key Responsibilities

- Architect and own the ML/AI stack for autonomous robotic systems, including model optimization pipelines, inference architecture, and integration interfaces.
- Evaluate and make key design trade-offs involving model accuracy, latency, power consumption, and edge versus cloud computation.
- Lead the deployment and optimization of ML models on edge devices, particularly NVIDIA Jetson platforms, ensuring production-grade reliability.
- Drive model optimization efforts including quantization, pruning, TensorRT compilation, memory management, and real-time performance tuning.
- Define architecture decisions between traditional machine learning approaches and modern deep learning or large-model techniques, selecting the most suitable solution for each robotics application.
- Provide technical leadership through architecture reviews, design guidance, and mentoring across ML stack components.
- Collaborate with systems and infrastructure teams on embedded Linux environments, containerization, CPU isolation, thermal management, and multi-process coordination.
- Develop and review production-quality C++ and Python code spanning research prototypes to deployed embedded applications.
- Ensure scalability, reliability, and maintainability of AI systems deployed on real robotic platforms.

Qualifications

Required

- Master's or PhD in Computer Science, Robotics, Artificial Intelligence,

Electrical Engineering, or a related discipline.

- Deep expertise in traditional machine learning algorithms and their application to system-level architectures.
- Strong experience with modern AI approaches, including deep neural networks, transformers, and large models, with sound architectural judgment for robotics applications.
- Extensive hands-on experience deploying ML models on embedded platforms, including quantization, hardware-specific optimization, memory constraints, and real-time inference.
- Proven ability to design and own complete ML architectures rather than isolated model components.
- Solid robotics fundamentals, including sensor integration, control loops, kinematics, and ROS/ROS 2.
- Strong proficiency in both C++ and Python for performance-critical robotics systems.
- Experience providing technical leadership, conducting design reviews, and mentoring engineering teams.
- Strong analytical and cross-functional collaboration skills.

Preferred

- Experience with robotics perception algorithms, including object detection, 3D scene understanding, SLAM, sensor fusion, and semantic segmentation.
- Hands-on experience optimizing perception models for edge hardware platforms.
- Familiarity with Docker, embedded Linux environments, and GPU acceleration technologies.
- Experience with large-scale AI infrastructure and distributed computing environments.
- Prior work with humanoid robots or other complex autonomous robotic systems.

About us:

ZEROGATE is a specialized provider of talent acquisition, talent outsourcing, and project delivery & management services, helping businesses achieve their goals through tailored, high-impact solutions. With expertise in sourcing globally recognized talent and managing complex projects, ZEROGATE supports clients with access to top-tier professionals, including technical consultants, PMOs, strategy consultants, and subject matter experts. By combining structured hiring processes, industry best practices, and client-focused project management, the company ensures efficient delivery, high-quality outcomes, and long-term partnerships built on trust, transparency, and measurable results.

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